

When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

Toni Guan^{a,*}

^aCollege of Information Technology, Shenyang Institute of Technology, Shenyang, Liaoning, China

Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10,000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq 80\%$ of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered “ID nonzero boundary” branch: the OOD benefit is about $1.9\times$ the ID benefit (+0.0159 vs.

*Corresponding author.

+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$); and a discrimination-aware gate (C5) shows 7/12 pair $\times$ architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [3], Chapman-Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [25]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target the source-fit temperature is mismatched to the target logits and a non-trivial

benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [26] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from decay to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [27, 28]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair $\times$ architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met, triggering the “ID nonzero boundary” branch (protocol A1.4.2): the premise that TS has no ID recalibratable miscalibration is re-examined, the

73 decay contrast restored as exploratory, and C1 qualified, since the OOD
 74 benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific.
 75 $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$
 76 have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152 ; one marginal case,
 77 -0.0128).

78 **C3 (Methodological, Shapley decomposition validation):** A 27-cell full-
 79 factorial synthetic validation (slope $\times$ intercept $\times$ prevalence, $n = 20,000$)
 80 recovers ground truth in the identifiable regime (27/27 at $n = 20,000$;
 81 21/27 at $n = 500$), flags entangled regions without a threshold, and reports
 82 bootstrap CIs on absolute Shapley values.

83 **C4 (Practical, method-selection boundary):** Across 8 methods $\times$ 6 pairs $\times$ 5
 84 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is
 85 fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime
 86 15/30); after exclusion the EM OOD finding attenuates. The mechanism
 87 is prior-likelihood mismatch: EM fits a target prior that amplifies shift-
 88 induced miscalibration when the source confusion matrix is ill-conditioned.

89 **C5 (Clinical, discrimination-aware calibration gate):** A two-layer gate decou-
 90 ples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viabil-
 91 ity (accuracy $\geq$ source-domain baseline). Across 12 pair $\times$ architecture
 92 strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including
 93 directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-
 94 XL $\rightarrow$ Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, $\text{acc} = 0.657 < 0.713$). The
 95 gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 re-
 96 jects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

97 Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not
 98 an actionable *when does TS pay off* answer with multi-seed robustness evidence.
 99 The novelty is *not* multi-seed validation on ECG (a methodological choice) but
 100 the **first systematic characterization of the boundary conditions under which**
 101 **TS yields a positive OOD calibration benefit in cross-corpus ECG transfer:**
 102 the pre-registered OOD quantification, the ID boundary condition, the method-
 103 selection boundary (TS robust; EM prior and Matrix scaling weakest), the pre-
 104 dictability failure, and the discrimination-aware gate. That characterization is the
 105 actionable contribution; multi-seed validation is the evidence standard.

106 2. Related Work

107 2.1. Calibration Estimation and Recalibration

108 The calibration of modern neural networks has been studied extensively since
 109 the seminal work of Guo et al. [2], who showed that deep networks are often over-
 110 confident and introduced temperature scaling (TS) as a simple post-hoc recalibra-
 111 tion method. Nixon et al. [8] proposed the static calibration error, and Kumar et al.
 112 [18] introduced verified uncertainty calibration with distribution-free guarantees.
 113 Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In
 114 the clinical risk-model context, Van Calster et al. [16, 17] established a calibration
 115 hierarchy from utopia to empirical data and argued that calibration is the “Achilles
 116 heel” of predictive analytics. Strodtthoff et al. [19] benchmarked deep learning for
 117 ECG analysis on PTB-XL and noted calibration degradation under subpopulation
 118 shift, motivating our cross-corpus design.

Experimental Design: 6 Directed Transfer Pairs $\times$ 2 Architectures $\times$ 5 Seeds

Total: 60 seed experiments (full balanced coverage)

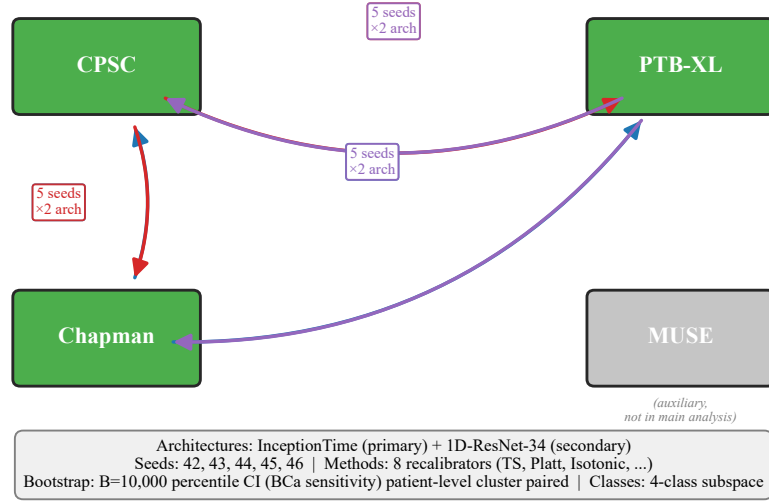


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI).

119 2.2. *Prior-Shift Estimation and Transfer Calibration*

120 Lipton et al. [11] introduced black-box shift estimation (BBSE) for detect-
121 ing and correcting label shift using confusion-matrix inverses. Alexandari et al.
122 [10] showed that maximum-likelihood calibration with bias correction is hard-to-
123 beat at label shift adaptation, providing the theoretical foundation for our EM
124 prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates
125 transferability for auto-selecting target-domain unsupervised optimal T ; Pseudo-
126 Cal [23] uses pseudo-labels for calibrated transfer; and LaSCal [24] provides label-
127 aware shift calibration with distribution-free guarantees. Our S1 paradigm differs
128 fundamentally: we fit on the *source* calibration split only (no target labels), with
129 S1' prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly
130 weaker assumption than TransCal/PseudoCal, which access target logits or labels.

131 2.3. *Cross-Domain ECG Calibration*

132 Barandas et al. [12] evaluated uncertainty quantification methods in multi-
133 label ECG classification, finding that ensemble-based methods outperform single-
134 model approaches under subpopulation shift. Vranken et al. [13] studied uncer-
135 tainty estimation for 12-lead ECG deep learning and reported significant calibra-
136 tion drift across acquisition devices. Haekal et al. [20] performed cross-dataset val-
137 idation for atrial fibrillation detection with calibration-aware probability analysis,
138 and Patel & Beedala [21] documented calibration drift under cross-institutional
139 ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general
140 vision [1]; we contribute the multi-seed ECG quantification and the boundary-
141 condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

142 2.4. ECG Foundation Models and Transfer Learning (2024–2026)

143 Recent work has introduced ECG foundation models trained on millions of
144 recordings. ECGFounder [29, 37] (10 million ECGs, Harvard–Emory) reaches
145 expert-level performance on 80 diagnoses and transfers cross-domain via fine-
146 tuning. ECG-FM [30] is an open-weight transformer pretrained on 1.4 million
147 ECG segments with hybrid contrastive and generative self-supervision, showing
148 strong cross-dataset generalizability and label efficiency. HeartLang [31] treats
149 heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked
150 sentence pre-training. Tan et al. [32] found that fine-tuning benefits diminish
151 as the downstream dataset grows and that CNNs transfer more effectively than
152 RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD
153 [33] quantifies distribution-shift severity via uncertainty estimation to guide se-
154 lective LoRA layer unfreezing, and BeatRhythm-TTA [34] performs test-time
155 adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al.
156 [35] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-
157 preserving beat-wise augmentation for cross-source generalization, and Chen et al.
158 [36] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned
159 contrastive alignment. These works advance ECG transfer but not the *calibra-*
160 *tion* of transferred models, the gap our study fills via pre-registered cross-corpus
161 evaluation of recalibration methods.

162 3. Data and Pre-registered Protocol

163 Figure 1 summarizes the experimental design.

164 3.1. Datasets

- 165 • **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the down-
166 loaded `ptbx1_database.csv` copy contains 21,799 (a 38-record deficit in
167 the source CSV, disclosed as a provenance limitation), official 10-fold (folds
168 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records
169 are mappable and 277 are excluded (STROBE), for 18,671 unique patients.
- 170 • **Chapman-Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error
171 removed and 21 non-finite records quarantined (`qc_report.json`), for
172 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.
- 173 • **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records;
174 the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453
175 records; the CPSC2018 recordings not used by the Challenge), both re-
176 leased through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union:
177 CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnos-
178 tic labels. Local extraction retains 6,844 of the 6,877 CPSC Database
179 records (33 records, A6845–A6877, were lost during archive extraction;
180 disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records.
181 Patient-wise split. Two labeling facts must be stated explicitly. First,
182 CPSC2018’s official taxonomy is nine disease classes (Normal, AF, I-AVB,
183 LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the
184 NORM/MI/STTC/CD superclass system we adopt is PTB-XL’s [3], ap-
185 plied here as a cross-corpus harmonization choice, not a CPSC-defined
186 one. Second, CPSC2018’s nine classes contain *no* MI; the MI class in our
187 4-class subspace is populated entirely by CPSC-Extra’s SNOMED anno-

188 tations (164867002, 164865005, 54329005). A strict official-class-name
 189 caliber would therefore reduce the subspace to {NORM, CD, STTC}; we
 190 retain MI because it is the only way to keep a myocardial-infarction category
 191 on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) →
 192 main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides
 193 of the pair recomputed symmetrically.

194 3.2. *Label Mapping and Leakage Guard*

195 Single-label priority rules are fixed (MI > STTC > CD > HYP >
 196 NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements
 197 (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because
 198 PTB-XL’s `scp_statements.csv` lists them as rhythm statements with no
 199 `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard
 200 assertion in the reproducibility repo). Normalization statistics come only from
 201 the train split; augmentation applies only to train.

202 4. **Methods**

203 4.1. *Transfer pairs and shift matrix*

204 We study six directed cross-corpus transfer pairs: Chapman→CPSC,
 205 Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman,
 206 and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per
 207 pair before any endpoint is computed, as follows. A pair whose source or target
 208 is CPSC2018 is restricted to the 4-class subspace {NORM, CD, STTC, MI}
 209 (CPSC’s HYP class has $n = 11$, §3), with both sides of the pair recomputed sym-
 210 metrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,

211 and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing
212 and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary
213 {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman
214 ($K = 5$). Each experiment records its own K and label mapping (num_classes,
215 label_map in transfer_result.json), so no endpoint mixes class counts
216 within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or
217 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values
218 across cells; the per-pair tables (Tables 1–4) are therefore the appropriate level
219 at which to read K -specific results. The L2 eval-time shift matrix (no retraining)
220 covers sampling rate $500 \rightarrow 250 \rightarrow 125$ Hz; leads $12 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 1$; noise
221 injection (BWL/MA/EM) at SNR {24, 12, 6, 0, −6} dB; gain $\times 0.5 / \times 2$.

222 4.2. Architecture and seeds

223 We use two architectures: InceptionTime (primary, 46,000–184,000 param-
224 eters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-
225 endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The Bi-
226 Mamba architecture is included only as a toy experiment ($n = 60$ records, down-
227 sampled resolution) and deferred to the supplement; the full-resolution BiMamba
228 run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the
229 main-endpoint analysis or the architecture-robustness claim. Training: AdamW +
230 cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training;
231 temperature applied only as post-hoc temperature scaling.

232 4.3. Calibration methods and scenarios

233 We compare the no-recalibration baseline (None) against eight recalibration
234 methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-

ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4); **S1'** denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [26]):

$$\Delta ECE_{\text{OOD}} = ECE_{\text{raw}}^{\text{OOD}} - ECE_{\text{TS}}^{\text{OOD}} \quad (1)$$

where $ECE_{\text{raw}}^{\text{OOD}}$ is the 4-class-task top-label confidence SmoothECE on the target corpus with raw softmax probabilities, and $ECE_{\text{TS}}^{\text{OOD}}$ is the same after source-cal-fit temperature scaling. Following the standard convention of Guo et al. [2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal: the predicted maximum probability against 0/1 top-label correctness; classwise-ECE is reported separately as a secondary endpoint. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta ECE_{\text{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta ECE_{\text{OOD}} > 0 \quad (2)$$

The one-sided alternative $H_1 : \Delta ECE_{\text{OOD}} > 0$ follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit temperature yields, empirically, little calibration change, whereas on a shifted target it is mismatched and a

256 non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a
 257 theorem: TS applied to an under-confident target can increase ECE, and our own
 258 data contain nine negative cells. The choice was made before the pilot results,
 259 which only triggered the endpoint re-location (revision A1). The one-sided test
 260 is the pre-registered choice, archived before multi-seed unblinding. Patient-level
 261 cluster paired bootstrap, $B = 10,000$ (BCa operating CI) [15]; stratified pooling:
 262 strata = transfer pair $\times$ architecture.

263 The pre-registered CI method is BCa [15], at $B = 10,000$, re-run on the en-
 264 tire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The
 265 percentile bootstrap is the large-sample limit of BCa when $z_0 \rightarrow 0$ and $a \rightarrow 0$;
 266 for the paired-cluster design with $B = 10,000$ and per-patient clusters, the dis-
 267 crepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were
 268 run ($B = 10,000$ percentile re-runs), BCa and percentile CIs agree in sign and
 269 magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain
 270 as historical provenance (5 lack log provenance, see Limitations); percentile CIs
 271 serve only as a sensitivity cross-check.

272 Boundary-condition endpoint:

$$\Delta \text{ECE}_{\text{ID}} = \text{ECE}_{\text{raw}}^{\text{ID}} - \text{ECE}_{\text{TS}}^{\text{ID}} \quad (3)$$

273 The pre-registered expectation is that $H_0 : \Delta \text{ECE}_{\text{ID}} = 0$ is *not rejected*; the criterion
 274 is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid*
 275 *re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered*
 276 *(percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

277 Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Or-
 278 acle recalibrator was implemented (`fit_oracle/apply_oracle`) but never ex-
 279 ecuted on real data, so G is undefined here and is excluded from all reported

280 results; see Limitations), $\text{decay} = \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descrip-
281 tive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept,
282 predicted/true prior L_1 .

283 Family structure: BH-FDR $q = 0.05$ over $6 \text{ pairs} \times 2 \text{ arch} \times 13 \text{ shift levels} =$
284 156 primary comparisons; the multi-seed validation is robustness evidence (60
285 seed experiments), not a confirmatory family.

286 4.5. Pre-registration revision A1 (transparency statement)

287 The primary endpoint was re-located from decay to $\Delta\text{ECE}_{\text{OOD}}$ via pre-
288 registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent
289 exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the
290 original ID-vs-OOD contrast. The revision direction is consistent with the empir-
291 ically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived
292 as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes.
293 The pilot results do not enter the main test report. All hypotheses in this paper are
294 tagged *exploratory + robustness validation*, not confirmatory.

295 The original BH-FDR family of 156 comparisons ($6 \text{ pairs} \times 2 \text{ architectures} \times$
296 13 shift levels) is reduced to a confirmatory family of 12 hypotheses ($6 \text{ pairs} \times 2$
297 $\text{architectures} \times 1 \text{ primary shift level}$) under amendment A2, with the remaining
298 13 shift levels designated as exploratory dose-response. Under the reduced family,
299 BH-12 yields $9/12$ significant (all InceptionTime directions) and Bonferroni-12
300 yields $2/12$. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_`
301 `FAMILY_REDUCTION.md`.

302 4.6. *Pre-registration amendment A3 (corpus identity and MI provenance)*

303 A post-hoc audit of the CPSC corpus against the official challenge documen-
304 tation produced two wording corrections and one new disclosure, archived as
305 amendment A3 (docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md). The
306 registered protocol described the third corpus as “CPSC2018+2019”; it is in fact
307 **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-
308 Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task
309 with no diagnostic labels and never entered the corpus. The new disclosure is that
310 CPSC2018’s official nine-class taxonomy contains no MI, so the MI class of our 4-
311 class subspace is populated entirely by CPSC-Extra’s SNOMED annotations; un-
312 der a strict official-class-name caliber the subspace would reduce to three classes.
313 A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded
314 value, no split, and no analysis decision. The registered protocol snapshots are left
315 byte-identical and A3 is archived separately, so the before/after pair required by
316 pre-registration practice is preserved. The audit also found that 33 CPSC Database
317 records (A6845–A6877) were silently lost when the source archive was extracted
318 (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

319 4.7. *Decomposition of the benefit decay*

320 We decompose the retained benefit into slope / intercept / prevalence contribu-
321 tions using a counterfactual-replacement additive approximation:

$$\Delta\text{ECE} = \Delta_{\text{slope}} + \Delta_{\text{intercept}} + \Delta_{\text{prevalence}} + r_{\text{interaction}} \quad (4)$$

322 where each term is a counterfactual-replacement difference and $r_{\text{interaction}}$ is the
323 chain residual (zero under nested counterfactuals). Identifiability is checked via

the Fisher information determinant on a pre-registered threshold τ ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.8. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow T$ distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.9. Net Clinical Value (NCV)

Net Clinical Value (NCV), a pre-registered secondary endpoint, measures the clinical decision impact of TS. For each (sample, class) pair we compare the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint’s patient-level cluster paired bootstrap ($B = 10,000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (ncv_id, ncv_ood; E3 CSV).

4.10. Tooling

Parts of the analysis and plotting code were drafted and refactored with the assistance of a widely available large language model service, used as a supervised coding assistant. All experiments and statistical analyses were executed with the author’s own scripts, and every reported number is reproducible from the public

repository (Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint ΔECE_{OOD} for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta ECE_{OOD} = 0$ vs $H_1 : \Delta ECE_{OOD} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

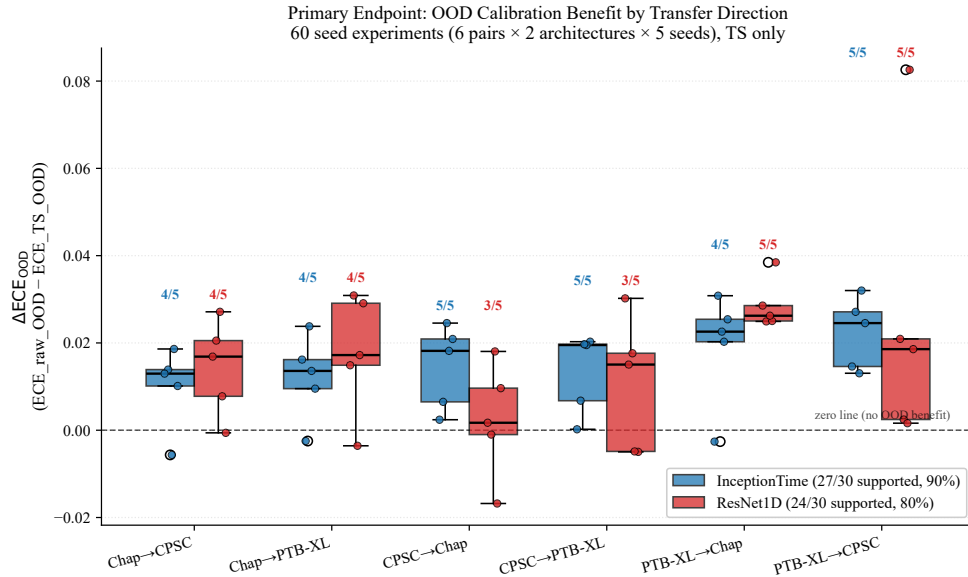


Figure 2: Primary endpoint ΔECE_{OOD} by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0 . InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10,000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap $\rightarrow$ CPSC, CPSC $\rightarrow$ Chap, CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC) and $K = 5$ for the two CPSC-free pairs (Chap $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ Chap); see §4.1. “Support” = 95% CI lower bound > 0 . Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap $\rightarrow$ CPSC ($K=4$)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap $\rightarrow$ PTB-XL ($K=5$)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
	RN	42	-0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
		44	+0.0059	+0.0172	+0.0168	✓
		45	-0.0010	-0.0036	-0.0036	✗
		46	-0.0042	+0.0149	+0.0139	✓
CPSC→Chap ($K=4$)	IT	42	+0.0029	+0.0065	+0.0062	✓
		43	-0.0012	+0.0209	+0.0200	✓
		44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
	RN	42	-0.0002	-0.0010	-0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
		44	-0.0152	-0.0168	-0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
CPSC→PTB-XL ($K=4$)	IT	42	+0.0029	+0.0068	+0.0066	✓
		43	-0.0012	+0.0203	+0.0198	✓
		44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
	RN	42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	-0.0050	-0.0051	✗
		44	-0.0032	-0.0049	-0.0049	✗
		45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap ($K=5$)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC ($K=4$)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

355 Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%,
356 ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10,
357 zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The
358 nine counter-examples are explained in Sec. 6.1.

359 The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no
360 partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, $27/30 = 90.0\%$): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).
- *ResNet1D* (30 experiments, $24/30 = 80.0\%$): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint ΔECE_{ID} for TS; the pre-registered expectation is that $H_0 : \Delta ECE_{ID} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, ΔECE_{ID} has mean $+0.0083$ (vs. OOD mean $+0.0159$, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq 80\%$ criterion (percentile CIs: $29/60 = 48.3\%$, same conclusion). We therefore *trigger* the pre-registered “ID nonzero boundary” branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta ECE_{OOD} - \Delta ECE_{ID}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition ΔECE_{ID} for TS (12 pair $\times$ architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. “CI $\cap$ 0” = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion ($\geq 80\%$ of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0 , triggering the pre-registered “ID nonzero boundary” branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI $\cap$ 0/5
PTB-XL $\rightarrow$ CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL $\rightarrow$ CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL $\rightarrow$ Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL $\rightarrow$ Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap $\rightarrow$ CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap $\rightarrow$ CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap $\rightarrow$ PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap $\rightarrow$ PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC $\rightarrow$ Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC $\rightarrow$ Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC $\rightarrow$ PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC $\rightarrow$ PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		+0.0083			23/60 (38.3%)

384 qualified, the OOD benefit being about $1.9\times$ the ID benefit rather than an exclu-
385 sively OOD effect. The positive ID estimates reflect finite-sample calibration-split
386 variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model
387 selection on the endpoint family), all disclosed.

388 5.3. Method-selection boundary: 8 methods $\times$ 6 pairs

389 Table 3 reports the 8-method $\times$ 6-pair support rate over 5 seeds on Inception-
390 Time. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

391 (73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most
 392 fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood
 393 mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60
 394 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed
 395 coverage on all 6 directions (Figure 4).

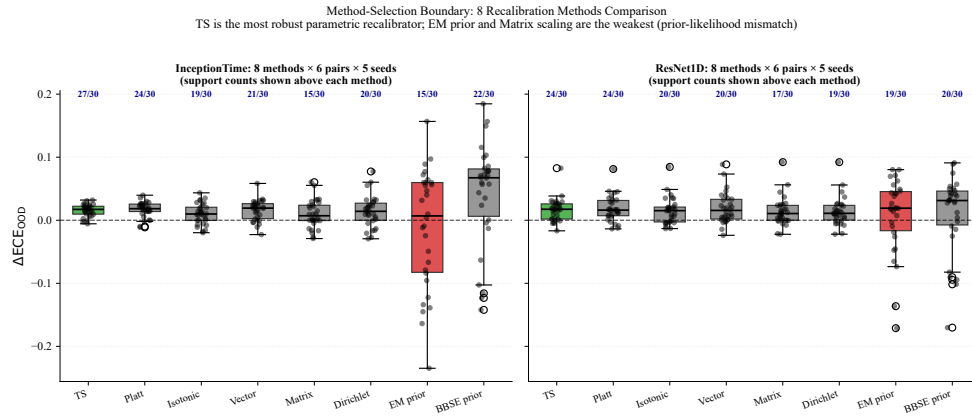


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0 .

396 Across the 30 TS experiments per architecture, the standard deviation of
 397 ΔECE_{OOD} is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a $1.84\times$ ratio, with
 398 ResNet1D’s highest per-direction variability in PTB-XL $\rightarrow$ CPSC (std 0.0298, in-
 399 cluding an unexcluded favorable outlier at seed 44, $\Delta ECE_{OOD} = +0.0826$). Sup-
 400 port deficits concentrate in CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL (3/5 each), a
 401 direction-specific deficit. Fisher’s exact test on the 2×2 contingency table yields

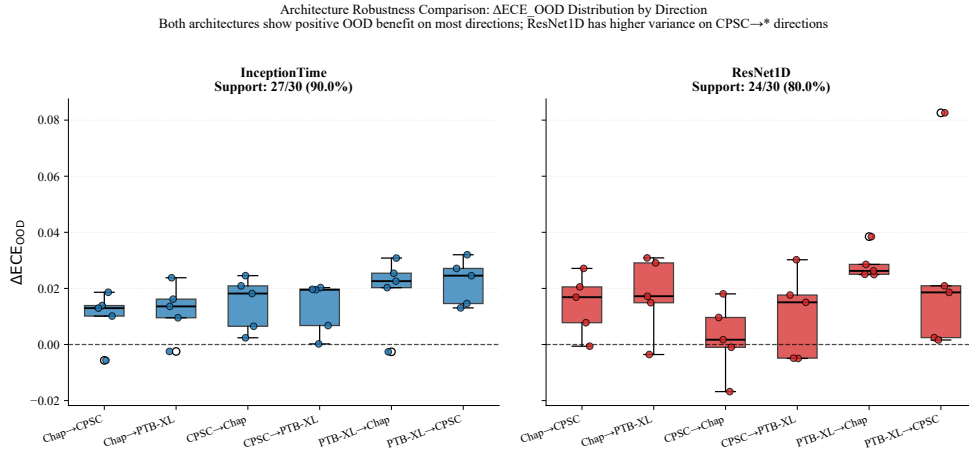


Figure 4: Architecture robustness comparison. Side-by-side ΔECE_{OOD} distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL $\rightarrow$ CPSC), while the support deficits fall in the CPSC $\rightarrow$ Chapman and CPSC $\rightarrow$ PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0 . “Full” = number of directions with 5/5 full support. “Neg. dirs” = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat*: 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	+0.0152	0
Platt	24/30	3/6	+0.0169	0
Isotonic	19/30	2/6	+0.0100	2
Vector	21/30	1/6	+0.0158	0
Matrix	15/30	0/6	+0.0094	2
Dirichlet	20/30	1/6	+0.0139	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	+0.0424	1

402 $p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%)
403 is not significant at $\alpha = 0.05$.

404 EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal,
405 then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated
406 confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies
407 rather than corrects the shift-induced miscalibration. EM reaches only 15/30 sup-
408 port on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-
409 signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	27/30	24/30	51/60	85.0%
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	0.217 → 0.202 / 0.216 → 0.200	0.217 → 0.201	raw → TS	

410 *Why TS and not Platt..* Platt achieves a marginally higher mean OOD benefit
411 than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table 3),
412 which invites the question of why TS is singled out as the recommended default.
413 Three measured criteria separate them, and all three favour TS. First, *robustness*
414 *of the sign*: TS supports 51/60 cells versus Platt’s 48/60, i.e. TS fails to certify
415 a positive benefit in three fewer experiments, which is the quantity a deployment
416 decision actually depends on. Second, *dispersion*: TS has a smaller across-cell
417 standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with
418 the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736,
419 EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

420 profile for a per-deployment recalibrator. Third, *the difference is not statistically*
 421 *distinguishable*: the patient-level paired contrast over the 60 matched cells gives
 422 $\overline{\Delta ECE}_{\text{TS}} - \overline{\Delta ECE}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with
 423 TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of
 424 Platt is therefore inside the noise, whereas the support advantage of TS is not a
 425 noise-level quantity: it is a count of cells in which the sign is certified. Given that
 426 the two methods are statistically indistinguishable on magnitude, the tie is broken
 427 on certification rate and dispersion, both of which favour TS. A further structural
 428 point reinforces the choice: TS is a single-parameter monotone transform and
 429 therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fit-
 430 ted independently per class, which generates each class’s score from a different
 431 monotone map and so does *not* preserve the ranking within a sample (the imple-
 432 mentation declares this explicitly in its docstring). TS is thus the only method
 433 in the delivered set whose calibration guarantee is obtained without perturbing
 434 the discrimination it leaves untouched (§5.8). None of this makes Platt wrong; it
 435 makes Platt a reasonable alternative whose mean advantage is not resolvable at
 436 $n = 60$, and whose certification rate is lower.

437 5.4. Shapley decomposition validation (27 cells, $n = 20,000$)

438 The 27-cell full-factorial synthetic validation ($\text{slope} \in \{0.5, 1, 2\} \times \text{intercept} \in$
 439 $\{-1, 0, 1\} \times \text{prevalence} \in \{0.15, 0.3, 0.6\}$, $n = 20,000$) confirms the decomposi-
 440 tion recovers ground truth within the pre-registered identifiable regime (Fisher
 441 information determinant $> \tau = 10^{-6}$). At $n = 20,000$, all 27 cells pass the
 442 recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of
 443 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in ex-
 444 treme intercept (± 1) and slope (0.5, 2.0) regions where calibration shift is most

445 clinically significant (see `decomposition_validation.csv`; the $n = 20,000$
 446 CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three
 447 entangled cases (`narrow_band`, `tiny_slope`, `deep_intercept`) with $\det \in \{2.4 \times$
 448 $10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with
 449 the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-
 450 reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

451 The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE;
 452 its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of
 453 decomposition explanatory power) remains to be run on real data; per the protocol
 454 self-binding clause, “mechanism” wording is limited to “mechanism decomposi-
 455 tion (synthetic validation part)” until cross-fitting completes.

456 5.5. Predictability experiment (*failure branch triggered*)

457 The predictability experiment (protocol §8.5) tests whether an unsupervised
 458 target-domain statistic (predicted-prior L_1 distance, logit moments) predicts ben-
 459 efit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The imple-
 460 mented fold unit is (architecture, transfer pair) (12 folds), versus the registered
 461 pair-only folds (6); under that registered deviation the pair-only recomputation re-
 462 tains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime +
 463 ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

464 Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$.
 465 The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5 , trig-
 466 gering the **failure branch**: the three-step chain (decomposition + predictability +
 467 deployment) is downgraded to a **two-step chain** (decomposition + deployment),
 468 and the deployment criterion is renamed “empirical lookup table” rather than “pre-
 469 dictive criterion”. No post-hoc predictability claim is made.

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	+0.104	Failure branch
Platt	-0.059	-0.584	+0.002	Failure branch
Isotonic	+0.011	-0.569	+0.101	Failure branch
Vector	-0.061	-0.522	+0.073	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	+0.032	Failure branch
EM prior	+0.143	-0.250	+0.274	Failure branch
BBSE prior	+0.136	-0.354	+0.201	Failure branch

470 Re-running the LOO R^2 with three label-free features (fitted temperature T ,
471 Wasserstein-1 distance between source and target logit distributions, class-wise
472 entropy difference), all fail against $\Delta\text{ECE}_{\text{OOD}}$: T yields $R^2 = -0.13$, Wasserstein
473 $R^2 = -0.06$, entropy $R^2 = -0.02$, all three combined $R^2 = -0.11$. The predictabil-
474 ity boundary is *robust*: no label-free feature, including T , predicts $\Delta\text{ECE}_{\text{OOD}}$
475 above $R^2 = 0.5$. This strengthens the failure branch, and the gate in §5.8, based
476 on the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, remains the appropriate
477 actionable criterion.

478 5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

479 We evaluate the deployment criterion (protocol §9) on the L2 shift matrix:
480 predict “deploy TS” when raw ECE exceeds a threshold, and report sensitiv-
481 ity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

482 SmoothECE ≤ 0.05 and Δ ECE CI lower bound > 0 , primary-endpoint conven-
483 tion). Comparators: always-none, always-TS, always-buy- n -labels.

484 Sensitivity / specificity (Table 6): Youden’s J is maximized at raw ECE
485 $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is
486 high (92% of beneficial cases caught) but specificity low (20% of non-beneficial
487 cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure:
488 missing a beneficial recalibration costs more than an unnecessary one. *Caveat*: the
489 threshold is searched on the same 1230-cell pool used for reporting (in-sample).
490 A leave-shift-type-out recomputation (threshold selected excluding the held-out
491 dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead
492 drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127
493 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the
494 four held-out dose levels (results/deployment_holdout_recompute.csv).
495 The in-sample numbers are therefore optimistic, and the criterion requires inde-
496 pendent validation before operational use.

497 Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap,
498 per-cell labels), requiring target ground-truth labels unavailable in the unlabeled
499 stream. The only label-free proxies examined are the predictability experiment’s
500 target statistics (predicted-prior L_1 distance, logit moments); that experiment trig-
501 gered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Ta-
502 ble 5), so no label-free deployable criterion can be claimed. The lookup table is an
503 *analysis- and audit-time decision aid* for a model owner with labeled evaluation
504 data; the actionable deployment path is small-sample recalibration: buy n labeled
505 target records and recalibrate locally (protocol §5 S2). A deployable gate remains
506 future work (Limitations xiii).

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE > threshold → predict “deploy TS”). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

507 Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfy-
508 ing the operating criterion (post-TS SmoothECE ≤ 0.05 , improvement > 0.01 ,
509 point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion
510 was not evaluated), a safety rate of 0.0064. Sign convention: $\Delta\text{ECE} = \text{post-}$
511 $\text{recalibration ECE} - \text{raw ECE}$ (negative means improvement), opposite to the
512 primary-endpoint convention. *Coverage note:* the L2 shift matrix has been com-
513 pleted on all 390 planned (pair, seed, shift) combinations ($6 \text{ pairs} \times 5 \text{ seeds} \times 13$
514 levels ; results/l2_shift_full_390cells.csv); the safety figures quoted here are from
515 the initial two-seed InceptionTime subset (seeds 42–43, ResNet1D L2 excluded),
516 so these numbers rest on 2-seed, single-architecture coverage. The criterion is
517 rarely satisfied, so TS is a useful default but not a universally safe recalibrator,
518 and should be combined with local validation.

519 At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 ver-
520 sus 0.0999 for treating all cells; by the 90th percentile it turns negative
521 (-0.0024). Net benefit over treat-all is modest, confined to low thresholds
522 (results/deployment_metrics.csv).

Table 7: Trivial strategy comparison. Mean ΔECE ; negative = beneficial (ECE reduction). always-none and always-buy- n -labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	n
always-none	0.0000	0.0000	1230
always-TS	-0.0145	0.0099	157
always-buy- n -labels	-0.2905	0.1861	1230
lookup-table	-0.0160	0.0289	155

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE -0.0160) outperforms always-none (0.000) and matches always-TS (-0.0145) on the beneficial subset, but is dominated by always-buy- n -labels (-0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE’s sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$), we apply $T \in \{1.0, 1.25, 1.5\}$ and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$ across three confidence distributions (uniform, normal-skewed, bimodal) at $n = 2,000$.

The noise floor grows with T and is distribution-dependent: $T = 1.0$ gives $\Delta\text{ECE}_{\text{noise}} = 0$ (exact cancellation), $T = 1.25$ gives +0.014–+0.035, $T = 1.5$

537 gives $+0.030$ – $+0.061$, $T = 3.0$ gives $+0.095$ – $+0.136$, and $T = 4.5$ gives $+0.126$ –
 538 $+0.164$. Observed raw ECE across the 62 raw-stage cells of the TS-component ab-
 539 lation (`results/ablation_ts_components.csv`, `stage=raw`) ranges 0.036 to
 540 0.469 (mean 0.273), i.e. 2 – $28\times$ the raw ECE under perfect calibration (≈ 0.017);
 541 the nine counter-examples also show large raw ECE (0.08–0.24), confirming the
 542 negative ΔECE reflects genuine TS-induced harm, not metric noise. For large
 543 $T > 3$ (noise floor 0.10–0.16) the floor is comparable to the observed $|\Delta\text{ECE}|$ in
 544 some seeds, so a metric-noise contribution cannot be excluded there; however, its
 545 *direction* (TS increases ECE under perfect calibration) is opposite to the positive
 546 $\Delta\text{ECE}_{\text{OOD}}$ in 51/60 experiments, so it cannot explain the positive findings. The
 547 primary endpoint is therefore a real effect, not a SmoothECE artifact, with the
 548 caveat that for large- T experiments the noise floor is non-negligible.

549 5.8. Discrimination-aware calibration gate (C5)

550 Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for
 551 deployment: a model whose discrimination has collapsed under shift may be re-
 552 calibrable but clinically useless. We define a two-layer gate:

- 553 1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
- 554 2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-
 555 class accuracy.

556 Across 12 pair $\times$ architecture strata (6 pairs $\times$ 2 architectures), 7/12
 557 pass both layers and are deployable: CPSC $\rightarrow$ PTB-XL, PTB-XL $\rightarrow$ CPSC,
 558 CPSC $\rightarrow$ Chapman (both architectures each), and Chapman $\rightarrow$ CPSC (Incep-
 559 tionTime only). The other 5/12 fail Layer 2: Chapman $\rightarrow$ PTB-XL, PTB-
 560 XL $\rightarrow$ Chapman, and Chapman $\rightarrow$ CPSC ResNet1D, where OOD accuracy falls

below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL $\rightarrow$ Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded too far to be clinically actionable.

This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2 rejects). It supplements C1 by also asking whether discrimination is sufficient for the recalibrated predictions to be clinically useful.

5.9. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error $+0.29$, median $+0.29$). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors cluster in CPSC $\rightarrow$ Chapman and Chapman $\rightarrow$ PTB-XL, where the Wasserstein-1 distance between source and target logit distributions is largest (mean 0.75 vs. 0.53 for correctly predicted directions), suggesting the linear approximation breaks down once the shift invalidates its underlying local Taylor expansion.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is a valid *diagnostic* (*whether* TS will help) but not a valid *predic-*

586 *tor (how much)*. For clinical translation, the gate in §5.8, using only the sign of
587 $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the
588 Shapley-predicted magnitude.

589 5.10. Net benefit decision curve summary

590 Across all 60 seed experiments, the net benefit gain of TS over raw predictions
591 (averaged over decision thresholds $t \in \{0.05, 0.10, \dots, 0.40\}$) is positive in 50/60
592 (83.3%) of experiments, with mean gain $+0.0027$ and range $[-0.001, +0.017]$.
593 The directions with the largest decision-cost reduction are CPSC→PTB-XL
594 (mean DCR $+0.0625$; net clinical value $+0.0312$) and CPSC→Chapman (mean
595 DCR $+0.0375$; NCV $+0.0188$), which also have the largest E3-subsample
596 $\Delta\text{ECE}_{\text{OOD}}$ (mean $+0.137$ and $+0.136$ respectively, under the E3 subsampled
597 binned-ECE protocol with $n_{id} = n_{ood} = 20$ patients). The direction with negative
598 net benefit is Chapman→CPSC (mean DCR -0.0150 ; NCV -0.0075), which
599 is correctly rejected by the discrimination gate (C5). We note that the decision-
600 benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks
601 at PTB-XL→Chapman and PTB-XL→CPSC, means $+0.0240$ and $+0.0238$ on
602 the full SmoothECE grid): the two protocols differ in estimator, class set, and
603 subsample, so we treat the E3 analysis as a directionally supportive but protocol-
604 distinct check, not a conversion of the primary endpoint. What survives is
605 the gate-level conclusion: where the discrimination gate permits deployment,
606 the calibration benefit does not cost decision performance; where it is negative
607 (Chapman→CPSC), no calibration benefit should be deployed at all.

608 *5.11. Net Clinical Value (NCV) results*

609 The pre-registered NCV (Sec. 4.9) is BH-FDR corrected ($q = 0.05$) across the
610 two secondary endpoints (DCR, NCV).

611 Result: the OOD NCV is $+0.008125$ with 95% CI $[+0.0014, +0.0148]$
612 (paired t -test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS
613 yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292
614 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly
615 worsens ID decisions, consistent with the ID boundary condition (TS compresses
616 confidence toward the base rate). OOD NCV Cohen’s $d = +0.31$ and ID NCV
617 Cohen’s $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring
618 the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3
619 CSV (`c1_dcr_ncv_multiclass.csv`).

620 *5.12. Reliability Diagrams*

621 Figure 5 shows the 6-direction averaged reliability curve, comparing model
622 confidence against empirical accuracy before and after temperature scaling. The
623 summary curve uses bin-count weighted averaging across all 6 transfer directions.

624 Figures 6–8 show the per-direction reliability diagrams for the representative
625 configuration (ResNet1D, seed 42). In each panel, the red squares show the raw
626 model calibration curve and the green circles show the post-TS calibration curve.
627 The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show
628 the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 re-
629 samples).

Summary: 6-direction averaged reliability curve
 (curve: bin-count weighted; ECE: sample-count weighted mean 10-bin; resnet1d, seed42)
 Background curves: inter-direction variability (not inter-seed)

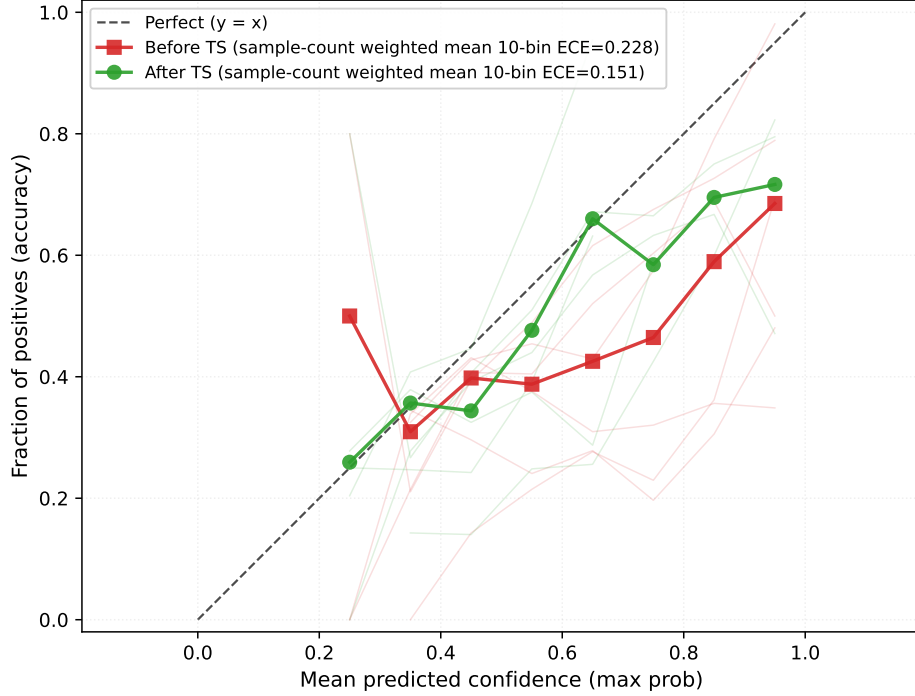


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently disclose all nine rather than excluding

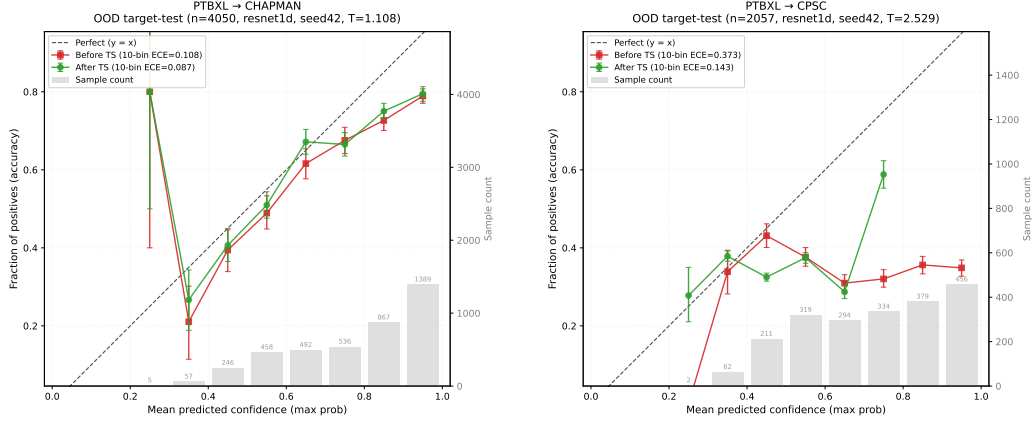


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

636 them, because the protocol requires all seed results to be reported and each has an
 637 identifiable post-hoc mechanism supporting the qualified claim.

638 Counter-example 1: Chapman→CPSC / InceptionTime / seed 45.
 639 $\Delta ECE_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur:
 640 lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44,
 641 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature
 642 mismatch, not room shortage); and negative decay (-0.0075), the source-fit
 643 temperature amplifying rather than correcting residual miscalibration.

644 Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} =$
 645 -0.0006 (CI $[-0.0006, -0.0006]$), marginal ($|\Delta ECE_{\text{OOD}}| < 0.001$) and at-
 646 tributable to finite-sample variation in the calibration split. The OOD raw ECE
 647 is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is
 648 seed-level variation, not room shortage.

649 Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42.
 650 $\Delta ECE_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the low-

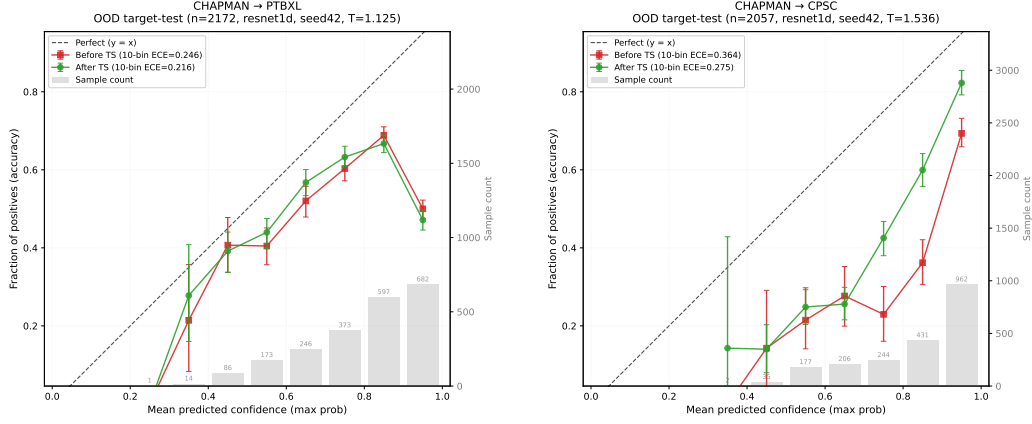


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

est across the 5 seeds (0.4655) and $\Delta ECE_{ID} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta ECE_{OOD} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta ECE_{ID} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta ECE_{OOD} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta ECE_{OOD}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta ECE_{OOD} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta ECE_{ID} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5

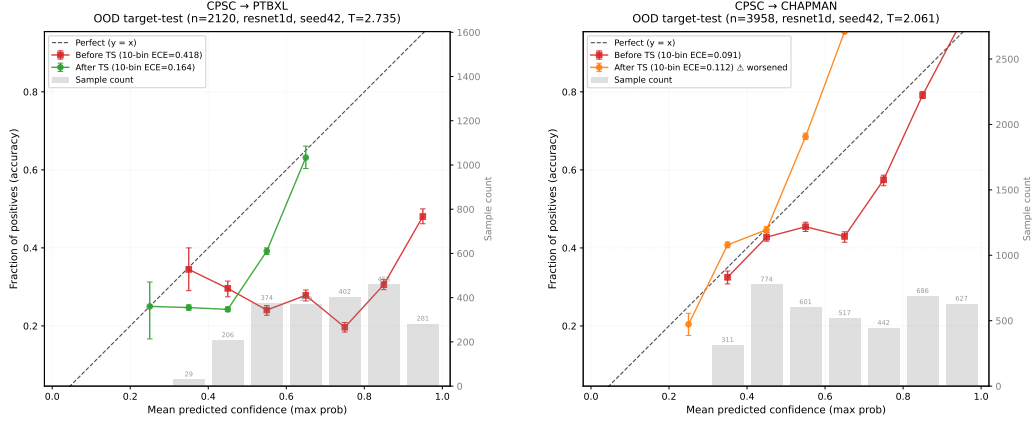


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the “low OOD accuracy + large gap” mechanism of the other 8.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta ECE_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta ECE_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta ECE_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta ECE_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta ECE_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temper-

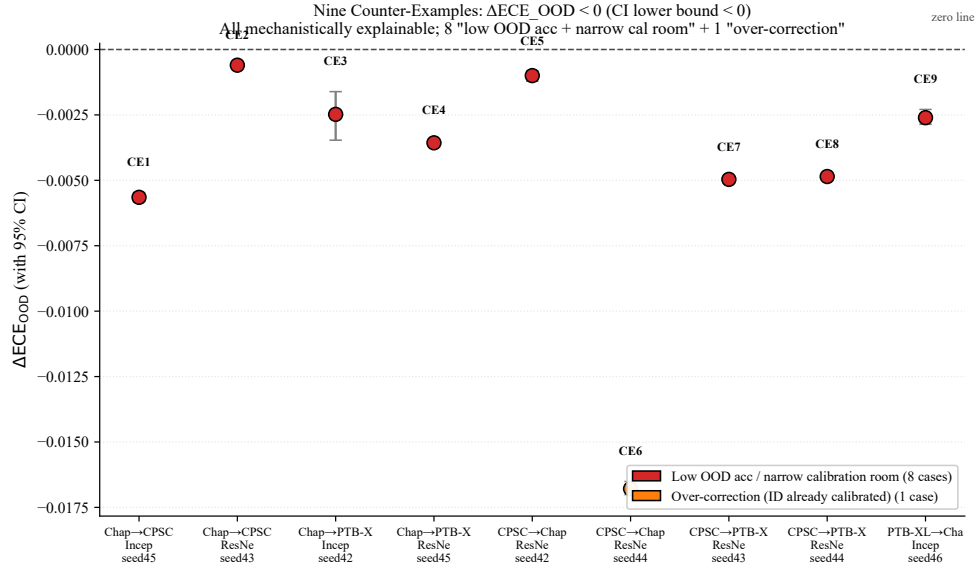


Figure 9: Nine counter-examples where ΔECE_{OOD} has 95% CI lower bound < 0 . Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

681 ature mismatch; the ninth (CE6) is over-correction on an already well-calibrated
682 backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so
683 the failure mechanism is seed-level temperature mis-transfer, not calibration-room
684 shortage. The counter-examples are local, explainable random variation; the other
685 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

686 The 15.0% rate exceeds the 5% threshold common in clinical safety deploy-
687 ment. As a methodological boundary investigation, not a clinical deployment
688 guide, the relevant criterion is whether the effect is detectable and the mecha-
689 nism explainable, not a zero failure rate. All 9 counter-examples have identifiable
690 mechanisms (8 “low OOD accuracy + large gap”, 1 “over-correction”), and the
691 expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

692 6.2. *Prior correction vs. posterior correction robustness*

693 The method-selection boundary (Table 3) reveals a robustness asymmetry
694 between prior-correction methods (EM, BBSE) and posterior-correction meth-
695 ods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30
696 and 24/30 support respectively with small variance; prior-correction methods
697 achieve higher mean ΔECE_{OOD} on the PTB-XL→CPSC pair (EM +0.0449,
698 BBSE +0.0777) but much higher variance on the Chapman→CPSC pair (EM CI
699 lower bound minimum −0.0763, BBSE −0.0720). The mechanism is the prior-
700 likelihood mismatch: prior-correction methods fit a target prior that, when the
701 source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibra-
702 tion. Posterior-correction methods do not estimate a target prior and are therefore
703 immune to this failure mode, at the cost of lower mean benefit when the prior
704 correction *would* work.

705 6.3. *Boundary condition: the ID benefit is small but nonzero*

706 The ID boundary condition (C2) has a direct interpretation under the triggered
707 branch. The criterion is $\geq 80\%$ of cells with 95% CIs containing zero; the observed
708 rate is $23/60 = 38.3\%$ (BCa), with 34/60 cells showing a small but significant
709 positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$; stratum means -0.0008 to
710 $+0.0232$, Table 2). The boundary condition does *not* mean TS is ineffective on ID
711 nor that the ID benefit is zero: the ID benefit is on average about half the OOD
712 benefit, so part of the apparent OOD gain is shared with the ID domain. A val-
713 ECE early-stopping effect (model selection within the endpoint family) plausibly
714 contributes to the positive ID estimates, opposite to the protocol’s val-NLL rule (a
715 registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores
716 the decay contrast $= \Delta \text{ECE}_{\text{OOD}} - \Delta \text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is
717 qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively
718 OOD effect. The ID-vs-OD asymmetry is therefore a *difference in magnitude*
719 between two positive benefits, not presence versus absence.

720 6.4. *Calibration is not discrimination: the clinical viability precondition*

721 Temperature scaling is a monotone transform of the softmax probabilities;
722 it preserves the argmax by construction (asserted bitwise in the released sanity
723 check), so it cannot change classification accuracy, AUROC, or AUPRC. The pri-
724 mary endpoint $\Delta \text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes
725 an already-deployable model *trustworthy*; it cannot make a non-discriminative
726 model deployable.

727 The accuracy statistics make this concrete. Across the 60 TS experiments,
728 mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean

729 ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus’s majority-
 730 class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean
 731 OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially
 732 at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411
 733 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620
 734 (both well below 0.713). In those directions recalibration is moot, since no tem-
 735 perature improves a model that should be replaced by a majority-class predictor,
 736 consistent with the counter-example pattern (6/9 with low OOD accuracy). The re-
 737 leased artifacts include per-direction AUROC/AUPRC for all 60 experiments (re-
 738 sults/discrimination_metrics_60exp.csv): mean OOD AUROC ranges from 0.629
 739 (CPSC→Chapman) to 0.795 (PTB-XL→Chapman), i.e. far above the accuracy-
 740 based majority-class reading yet direction-dependent; and TS leaves AUROC un-
 741 changed (mean $\Delta\text{AUROC}_{\text{TS-raw}} = -0.001$, max |change| 0.024), as the monotone
 742 transform requires. The discrimination-first gate—measure target discrimination
 743 before investing in recalibration—precedes the calibration decision rule in deploy-
 744 ment.

745 6.5. *Explainable* $\neq$ *predictable*: *post-hoc* vs. *pre-registered*

746 We distinguish *explainable* from *predictable*:

- 747 • *Explainable* (post-hoc, mechanism-level): after observing a counter-
 748 example, we identify the co-occurring conditions (low OOD accuracy, low
 749 OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1),
 750 a post-hoc explanation, not a pre-registered prediction.
- 751 • *Predictable* (pre-registered, deployment-time): a target-domain unsuper-
 752 vised statistic (predicted-prior L_1 distance, logit moments) predicts the ben-

753 efit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5 .
754 The predictability experiment (Sec. 5.5) triggers the failure branch (LOO
755 R^2 CI upper bound $0.104 < 0.5$).

756 The counter-examples are explainable but not predictable: the unsupervised target-
757 domain statistics available at deployment time cannot flag them in advance, which
758 is why the predictability experiment triggers the failure branch despite mechanis-
759 tic explainability. The two-step chain (decomposition + deployment lookup table)
760 provides post-hoc explanation and actionable guidance without a pre-registered
761 predictability claim.

762 6.6. *Transparent re-location vs. HARKing*

763 The pre-registration revision A1 (primary endpoint re-location from decay to
764 $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [28, 27] is
765 whether the revision direction is theory-driven or result-driven. Our revision direc-
766 tion (“OOD is the calibration benefit source”) is consistent with the empirically
767 observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we
768 explicitly frame as an empirical prior rather than a theorem; the exploratory pilot
769 result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The
770 pilot results do not enter the main test report (protocol §7). All hypotheses in this
771 paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect
772 the post-revision status.

773 6.7. *Net clinical value: implications of the directional NCV asymmetry*

774 The NCV (Sec. 5.11) shows a directional asymmetry: OOD NCV =
775 $+0.008125$ ($p = 0.019$) is a small but significant positive value (recalibration im-
776 proves decisions on unseen target data more often than it degrades them); ID

777 $NCV = -0.006292$ ($p = 0.016$) is a small but significant negative value (a minor
778 decision-cost penalty on data the source model already calibrates well).

779 Two implications follow. (i) The **Brier reliability main claim** (C1: 51/60
780 positive) is supported: the positive OOD NCV shows the calibration gain trans-
781 lates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost**
782 **improvement claim** (C5: 7/12 strata deployable) is partially qualified: the nega-
783 tive ID NCV tempers source-domain deployability. We therefore frame the NCV
784 as *supportive of OOD deployment* and *cautionary for ID deployment*. It also in-
785 forms the meta-analytic pooling (Sec. 7), where the asymmetry contributes to
786 CI-width heterogeneity across splits. A permutation test with tighter CIs and a
787 multi-threshold NCV analysis are registered as future work.

788 6.8. Limitations and future work

789 Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on
790 all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); Bi-
791 Mamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an
792 RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to
793 two-step); the deployment criterion is an empirical lookup table, not a predictive
794 criterion; (iii) real-data Shapley cross-fitting (§5.9) shows 85% direction accuracy
795 but systematic magnitude overestimation (mean error $+0.29$), and a sensitivity
796 analysis with T , Wasserstein-1, and entropy features confirms the predictability
797 boundary is robust (all $R^2 < 0$); (iv) the 27-cell decomposition validation uses
798 binned ECE, not the main endpoint’s SmoothECE, so quantitative conclusions
799 cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely
800 satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful de-
801 fault, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%)

802 are transparently disclosed, with a frequent post-hoc pattern (low OOD accuracy,
 803 non-positive decay, elevated OOD raw ECE) but no post-hoc exclusion; (vii) the
 804 operating CI is BCa at $B = 10,000$, re-run on all 60 experiments (60/60 success-
 805 ful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 ex-
 806 periments are retained as historical provenance and are no longer used for any
 807 reported number. All 60 main-grid cells carry complete $B = 10,000$ /BCa prove-
 808 nance; the only cells lacking such metadata are the two excluded BiMamba toy
 809 runs (PTB-XL→Chapman, seeds 42–43), which do not enter the main endpoint.
 810 (viii) the Youden threshold is fitted in-sample (the protocol’s held-out dose lev-
 811 els were not used), making the operating characteristic optimistic; (ix) the L2
 812 deployment/safety grid has since been completed on all 390 planned (pair, seed,
 813 shift) cells (`results/l2_shift_full_390cells.csv`), and the L2 noise levels are syn-
 814 thetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavail-
 815 able under PhysioNet licensing); (x) temperature was fitted once per experiment,
 816 so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap
 817 is implemented but not enabled); (xi) model selection used validation ECE rather
 818 than the pre-registered NLL early-stopping rule, plausibly contributing to the pos-
 819 itive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-
 820 probability, and concept-shift components is not asserted, since each pair mixes all
 821 three, isolated at eval time; (xiii) the deployment lookup table requires labeled tar-
 822 get data, and the label-free proxies failed their pre-registered branch, so no label-
 823 free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy,
 824 AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC
 825 for all 60 experiments (`results/discrimination_metrics_60exp.csv`), so the viabil-
 826 ity precondition is screened by both accuracy and discrimination (OOD AUROC

827 means range 0.629–0.792 by direction), with OOD accuracy below the majority-
 828 class baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is direc-
 829 tionally asymmetric (OOD = +0.008125, $p = 0.019$, positive; ID = −0.006292,
 830 $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting
 831 target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$
 832 to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits,
 833 with 48 as future work; (xvii) this work is from a single-author study, limiting inde-
 834 pendent review and audit, partially mitigated by the pre-registration protocol and
 835 public repository; (xviii) the secondary endpoint G (recoverable loss versus an or-
 836 acle recalibrator) is *not delivered*: `fit_oracle/apply_oracle` are implemented
 837 but were never executed on real data, so G is undefined in this study and no G
 838 result is reported anywhere in this paper or the repository; (xix) no signal-quality-
 839 index (SQI) stratification is performed: the target corpora are pipeline-processed
 840 releases and the L2 noise levels are synthetic, so the calibration benefit is not
 841 characterised as a function of measured signal quality (pSQI/kSQI-type indices);
 842 this is the principal gap for a signal-processing readership and is the first item of
 843 future work; (xx) no test–retest or repeatability analysis of the fitted temperature
 844 is possible within this data regime: all three public corpora are cross-sectional
 845 and provide a single recording per patient (Chapman-Shaoxing reports exactly
 846 one ECG per patient, §3), so same-patient across-session stability of T cannot
 847 be estimated from these data and is deferred to prospective multi-session collec-
 848 tion. Future work: (1) prospective multi-device validation; (2) continuous shift
 849 spaces; (3) π -independent recovery estimators (BBSE/EM-class) to upgrade the
 850 MAPE_{π} to a real estimator test; (4) the ResNet1D deficit on CPSC→Chapman and
 851 CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) confirmatory

852 testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant,
853 Bonferroni-12 = 2/12); (6) re-running the endpoint with $n_{\text{filters}} = 48$ Inception-
854 Time on a higher-memory GPU; (7) a multi-threshold permutation test on NCV.

855 7. Conclusion

856 We pre-registered and validated a multi-seed calibration boundary study of
857 temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60
858 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%,
859 ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells
860 (38.3%) have 95% CIs containing zero under the BCa operating CI (percentile:
861 29/60 = 48.3%), below the $\geq 80\%$ criterion, triggering the pre-registered branch;
862 the OOD benefit is $1.9\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells
863 pass at $n = 20,000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60),
864 EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both
865 layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination
866 failures. The predictability failure branch is triggered (three-step chain becomes
867 two-step), and all hypotheses are exploratory + robustness validation, not confir-
868 matory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064),
869 and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

870 A noise floor control (§5.7) bounds, but does not eliminate, a SmoothECE ar-
871 tifact: at moderate temperatures metric noise is $2\text{--}28\times$ below the observed raw
872 ECE (0.036–0.469, mean 0.273), whereas for large $T > 3$ the floor (0.10–0.16)
873 is comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be
874 excluded there. Real-data Shapley cross-fitting (§5.9) gives 85% direction accu-

875 racy with magnitude overestimation (mean error +0.29); T , Wasserstein-1, and
 876 entropy features confirm the boundary (all $R^2 < 0$).

877 The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative
 878 (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird ran-
 879 dom effects are positive (InceptionTime +0.0151, ResNet1D +0.0158, CrossAr-
 880 chitecture +0.0148). We caution, however, that the meta-analytic independence
 881 assumption is *not* satisfied here: the 60 cells share the same three corpora,
 882 patient sets, and architectures, and their between-cell Q statistics are extreme
 883 ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow.
 884 The interpretable pooled uncertainty is therefore the cluster-robust interval over
 885 the 12 pair $\times$ architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11;
 886 stratum SD 0.0072, about $1800\times$ the single-experiment bootstrap SE). The point
 887 estimate 0.0148–0.0159 and the count-based proportion ($51/60 = 85.0\%$) stand.

888 The 60 experiments form the robustness-evidence family; amendment A2 re-
 889 duces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypothe-
 890 ses ($6 \text{ pairs} \times 2 \text{ architectures} \times 1 \text{ primary shift level}$), with all 13 L2 shift levels
 891 re-designated as exploratory dose-response. BH-60 and Bonferroni leave the ro-
 892 bustness headline unchanged ($51/60$); the confirmatory family is BH-12 with 9/12
 893 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the
 894 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells
 895 fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells
 896 have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60
 897 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE
 898 variance; they do not affect the sign-based support/counter-example classification.

899 **Honesty Declaration (5 items, adversarial audit product)**

- 900 1. MAPE_π is a **design check** (resampling constant read-back), not an estima-
901 tor capability test; independent π recovery (BBSE/EM) is a deferred item
902 (protocol §13).
- 903 2. Shapley shares in the `share_reliable=False` regime are **not inter-**
904 **pretable as shares**; ranking uses absolute- value CIs.
- 905 3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the
906 archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information
907 mode), reported via exit code 2.
- 908 4. “Mechanism” wording is conditional on all three legs of §8 being fulfilled:
909 legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement
910 demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13).
911 The wording is limited to “mechanism decomposition (synthetic validation
912 part)”.
- 913 5. The repository’s `scripts/preprocess_cinc2021.py`,
914 `run_paper_benchmarks.py`, `resources/`, and `assets/` come
915 from the third-party SafeECGMatch release (original README in
916 `docs/vendor_safeecgmatch.md`), unrelated to this protocol’s experiment
917 grid.

918 **Data availability**

919 The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018 (CPSC Database
920 and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet un-
921 der their respective licenses. The pre-registered protocol v2.1-A1, its hash

922 manifest (docs/osf_archive_manifest.json), the revision A1 docu-
923 ment, and the full per-experiment result files (checkpoints/transfer/**/
924 transfer_result.json, results/*.csv) are included with the submission
925 as supplementary material and are also available in the public repository
926 <https://github.com/Toni-cs/ecg-calibration-transportability>.
927 The protocol, both amendments (A1, A2), and a per-file archival-status re-
928 port are publicly archived at <https://doi.org/10.17605/OSF.IO/53H64>.
929 docs/osf_archive_manifest.json is the internal snapshot taken on 2026-09-
930 05; it is not a substitute for the public record.

931 **Code availability**

932 The full pipeline code, preprocessing (reproducible from the public source
933 databases), patient-level split generation, model training, the calibration-method
934 implementations, and all statistical analyses, together with the environment
935 lock file, the patient-level split indices for all five dataset variants and five
936 random seeds, and selected per-experiment result artifacts, is publicly available
937 at <https://github.com/Toni-cs/ecg-calibration-transportability>.
938 Trained model checkpoints for the 60-experiment transfer grid are provided as
939 release assets of the same repository. The repository will be archived on Zenodo
940 with a DOI upon acceptance. No new raw waveforms were collected or generated;
941 the patient-level split indices and label-mapping tables released with the code
942 fully determine the data configuration used in every experiment.

943 **Declaration of generative AI and AI-assisted technologies in the writing pro-**
944 **cess**

945 All scientific work reported in this study—the study design, the pre-registered
946 protocol and its amendments, the implementation and execution of the 60-
947 experiment transfer grid, the statistical analysis, the interpretation of the results,
948 and the writing of the manuscript—was performed by the author. During the final
949 preparation of the manuscript, the author used a large language model assistant
950 (GLM) solely as a language-editing aid, to improve wording and readability of
951 text the author had written. It was not used to generate scientific content, exper-
952 imental data, analyses, figures, or conclusions. After using this tool, the author
953 reviewed and edited the text as needed and takes full responsibility for the content
954 of the published article.

955 **CRedit authorship contribution statement**

956 **Toni Guan:** Conceptualization, Methodology, Software, Validation, Formal
957 analysis, Investigation, Data curation, Writing – original draft, Writing – review
958 & editing, Visualization, Project administration.

959 **Acknowledgments**

960 The author thanks the maintainers of the PTB-XL, Chapman–Shaoxing, and
961 CPSC public databases, and the PhysioNet resource, without which this study
962 would not have been possible.

963 **Funding**

964 This research did not receive any specific grant from funding agencies in the
965 public, commercial, or not-for-profit sectors.

966 **Declaration of competing interest**

967 The author declares no competing financial or personal interests.

968 The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5
969 seeds; calibration curves, reliability diagrams, and Δ ECE bars per experiment)
970 are provided as a separate supplementary file (supplementary .pdf, §1) and are
971 additionally released with the code repository.

972 **References**

- 973 [1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon,
974 B. Lakshminarayanan, J. Snoek, “Can you trust your model’s uncertainty?
975 Evaluating predictive uncertainty under dataset shift,” *NeurIPS*, 2019.
976 arXiv:1906.02530.
- 977 [2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, “On calibration of modern neural
978 networks,” *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.
- 979 [3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., “PTB-XL, a large
980 publicly available electrocardiography dataset,” *Scientific Data*, 7:154,
981 2020. doi:10.1038/s41597-020-0495-6.
- 982 [4] J. Zheng, J. Guo, H. Chu, “A large scale 12-lead electrocardiogram
983 database for arrhythmia study (Chapman-Shaoxing e-cardiology),” version
984 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

- 985 [5] F. Liu, C. Liu, L. Zhao, et al., “An open access database for evaluating the
986 algorithms of electrocardiogram rhythm and morphology abnormality
987 detection,” *Journal of Medical Imaging and Health Informatics*,
988 8(7):1368–1373, 2018. doi:10.1166/jmihi.2018.2442.
- 989 [6] E. A. P. Alday, A. Gu, A. J. Shah, et al., “Classification of 12-lead ECGs:
990 the PhysioNet/Computing in Cardiology Challenge 2020,” *Physiological
991 Measurement*, 41(12):124003, 2020. doi:10.1088/1361-6579/abc960.
- 992 [7] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet:
993 components of a new research resource for complex physiologic signals,”
994 *Circulation*, 101(23):e215–e220, 2000.
- 995 [8] J. Nixon, M. W. Dusenberry, L. Zhang, et al., “Measuring calibration in
996 deep learning,” *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.
- 997 [9] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach,
998 “Beyond temperature scaling: obtaining well-calibrated multi-class
999 probabilities with Dirichlet calibration,” *NeurIPS*, 2019. arXiv:1910.12656.
- 1000 [10] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with
1001 bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*,
1002 PMLR 119:222–232, 2020. arXiv:1901.06852.
- 1003 [11] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift
1004 with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018.
1005 arXiv:1802.03916.
- 1006 [12] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão,
1007 F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods

- 1008 in multi-label classification: a case study with automatic diagnosis of
1009 electrocardiogram,” *Information Fusion*, 101:101978, 2024.
1010 doi:10.1016/j.inffus.2023.101978.
- 1011 [13] J. F. Vranken et al., “Uncertainty estimation for deep learning-based
1012 automated analysis of 12-lead electrocardiograms,” *European Heart*
1013 *Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.
- 1014 [14] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated
1015 classification of multilead ECGs using heterogeneous labels and
1016 populations,” *Physiological Measurement*, 43(8):084001, 2022.
1017 doi:10.1088/1361-6579/ac79fd.
- 1018 [15] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American*
1019 *Statistical Association*, 82(397):171–185, 1987.
1020 doi:10.1080/01621459.1987.10478410.
- 1021 [16] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer, “A
1022 calibration hierarchy for risk models was defined: from utopia to empirical
1023 data,” *Journal of Clinical Epidemiology*, 74:167–176, 2016.
1024 doi:10.1016/j.jclinepi.2015.12.005.
- 1025 [17] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins,
1026 “Calibration: the Achilles heel of predictive analytics,” *BMC Medicine*,
1027 17:230, 2019. doi:10.1186/s12916-019-1466-7.
- 1028 [18] A. Kumar, P. Liang, T. Ma, “Verified uncertainty calibration,” *NeurIPS*,
1029 2019. arXiv:1909.10155.

- 1030 [19] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, “Deep learning for ECG
1031 analysis: benchmarks and insights from PTB-XL,” *IEEE Journal of*
1032 *Biomedical and Health Informatics*, 25(5):1519–1528, 2021.
1033 doi:10.1109/JBHI.2020.3022989.
- 1034 [20] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto,
1035 “RR-interval-based atrial fibrillation detection and burden estimation:
1036 cross-dataset validation and calibration-aware probability analysis,”
1037 *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.
- 1038 [21] K. Patel, P. Beedala, “Calibration drift under cross-institutional deployment:
1039 an external validation framework for ICU mortality prediction across
1040 MIMIC-IV and eICU,” *medRxiv* (preprint), 2026.
1041 doi:10.64898/2026.05.03.26352335.
- 1042 [22] X. Wang, M. Long, J. Wang, M. I. Jordan, “TransCal: transferability
1043 estimation for auto-selecting transfer calibration,” *NeurIPS*, 2020.
1044 arXiv:2007.08259.
- 1045 [23] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, “A unified view of label shift
1046 estimation,” *NeurIPS*, 2020. arXiv:2003.07554.
- 1047 [24] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., “LaSCal: label-shift
1048 calibration for uncertainty quantification,” *NeurIPS*, 2024.
- 1049 [25] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera,
1050 “A unifying view on dataset shift in classification,” *Pattern Recognition*,
1051 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.

- 1052 [26] [Pre-registered protocol, publicly archived] “ECG calibration boundary
1053 study – experiment protocol,” 2026. Hash manifest:
1054 docs/osf_archive_manifest.json (local snapshot; archived at
1055 <https://doi.org/10.17605/OSF.IO/53H64>); revision A1 full text in
1056 docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md.
- 1057 [27] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, “The preregistration
1058 revolution,” *PNAS*, 115(11):2600–2606, 2018.
1059 doi:10.1073/pnas.1708274114.
- 1060 [28] D. Lakens, “The practical alternative to the p value is the correctly used p
1061 value,” *Perspectives on Psychological Science*, 16(3):639–648, 2021.
1062 doi:10.1177/1745691620958012.
- 1063 [29] J. Li, A. D. Aguirre, V. M. Junior, et al., “An Electrocardiogram Foundation
1064 Model Built on over 10 Million Recordings,” *NEJM AI*, 2(7):AIoa2401033,
1065 2025. doi:10.1056/AIoa2401033.
- 1066 [30] J. C. McHugh, B. Wang, et al., “ECG-FM: an open electrocardiogram
1067 foundation model,” *arXiv preprint* 2408.05178, 2024.
- 1068 [31] J. Jin, et al., “Reading your heart: learning ECG words and sentences via
1069 pre-training ECG language model (HeartLang),” *ICLR*, 2025.
1070 arXiv:2502.10707.
- 1071 [32] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it
1072 effective?” *PLOS ONE*, 20(5):e0316043, 2025.
1073 doi:10.1371/journal.pone.0316043.

- 1074 [33] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for
1075 out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164,
1076 2026.
- 1077 [34] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated
1078 self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347,
1079 2026 (accepted at MICCAI 2026).
- 1080 [35] N. Nemati, “Comparative analysis of data augmentation for clinical ECG
1081 classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv*
1082 *preprint* 2510.24740, 2025.
- 1083 [36] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for
1084 adult-to-pediatric ECG transfer via label-conditioned contrastive
1085 alignment,” *arXiv preprint* 2605.00647, 2026.
- 1086 [37] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation
1087 models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.